



PORTFOLIO CASE STUDY

NBA Analytics Platform

Six versioned sources become a clear view of team performance, context and change.



01 · HISTORY

Historical performance and scoring

02 · CONTEXT

Home context, consistency and shooting

03 · PROFILES

Offense and historical player profiles

04 · PEAK

Historical peak vs. 2013–2022

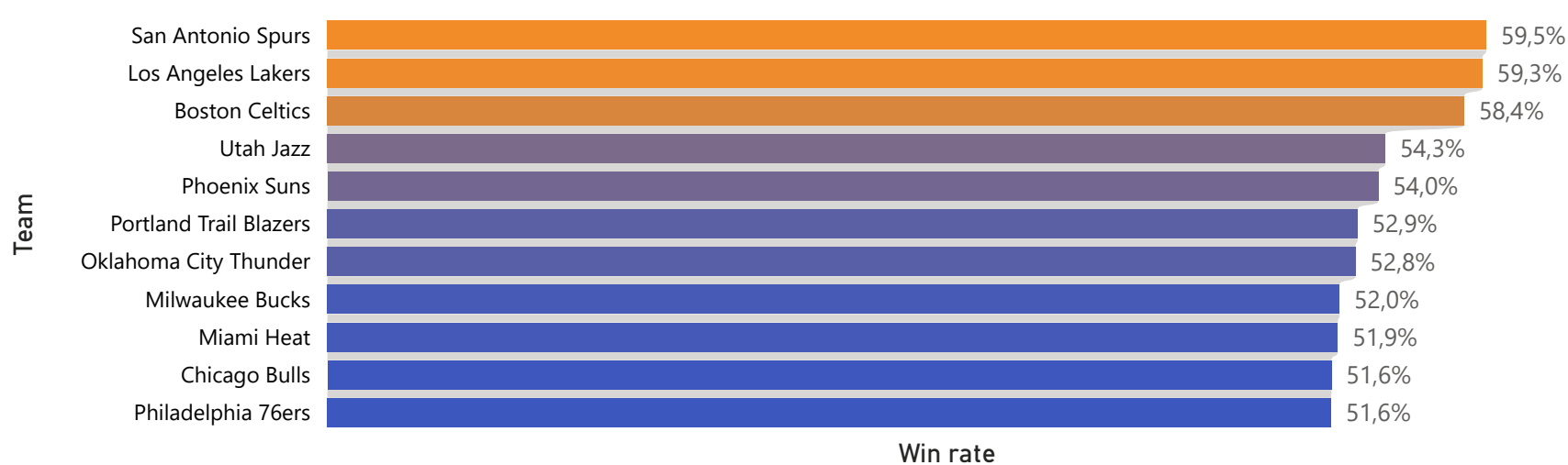
6 VERSIONED CSVs · **161,111 SOURCE ROWS** · **65,642 GAMES** · **13 TESTS**

Python ETL → SQL Server → 15 analytical objects → Power BI DirectQuery



QUESTION Which team led the historical win-rate ranking, and how did scoring change by decade?

Historical win rate | seasons 1946–2022



Team

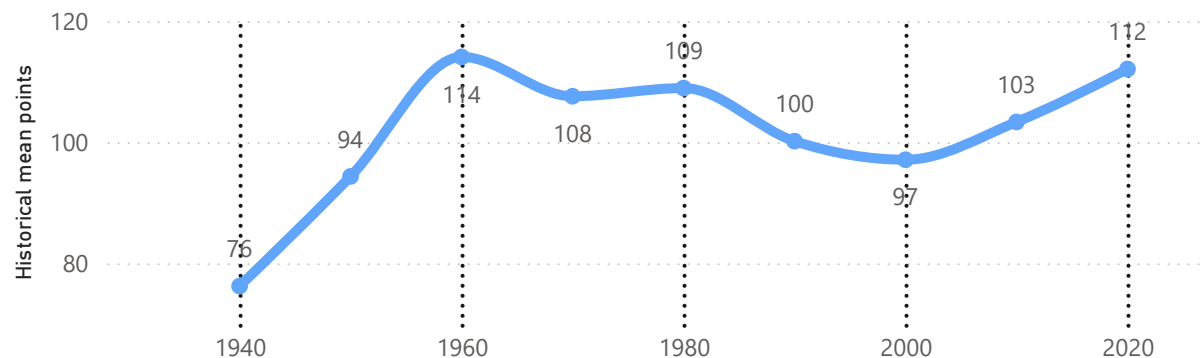
Todas

KEY FINDING · FULL SAMPLE

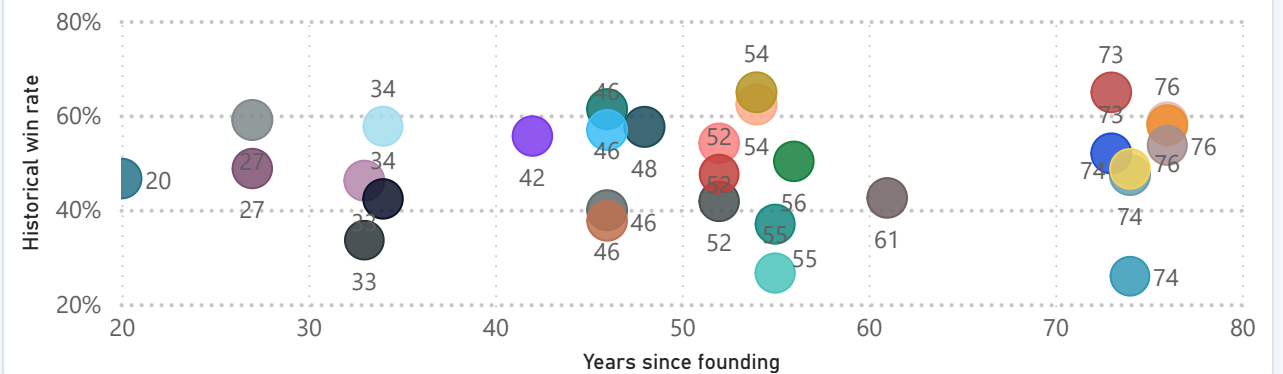
The Spurs lead the historical win rate: 59.53% across 4,077 games.

Eligible team-game scoring rises from 76.23 PPG in the 1940s to 112.07 in the 2020s.

Scoring by season decade | points per game (PPG)



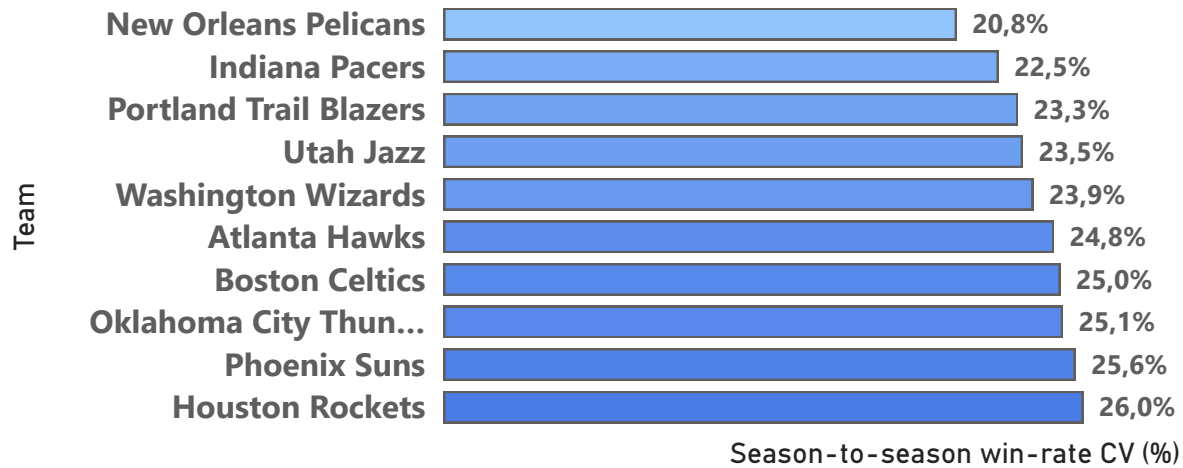
Franchise age vs. win rate | seasons 2020–2022





QUESTION How does home court relate to scoring, and which teams vary less across seasons?

Season-to-season consistency | 1946–2022 | win-rate CV



Season decade

Todas

Team

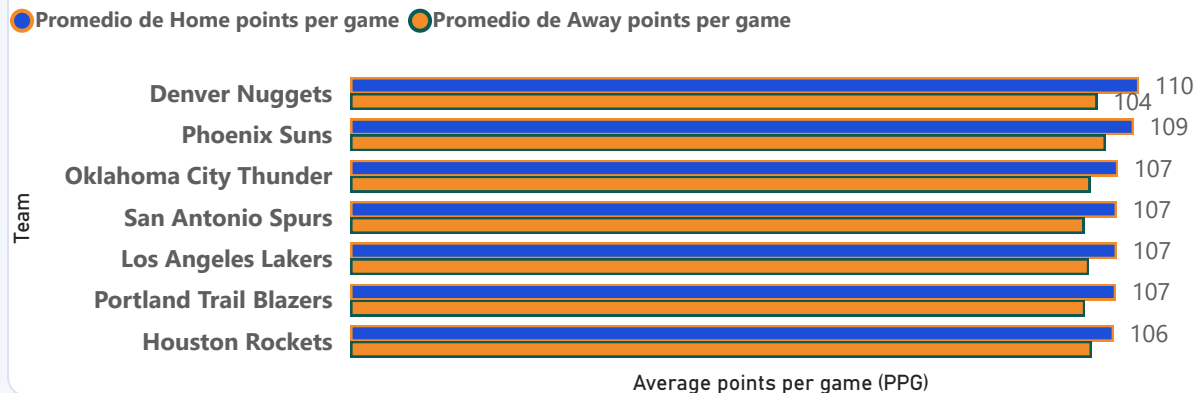
Todas

KEY FINDING · FULL SAMPLE

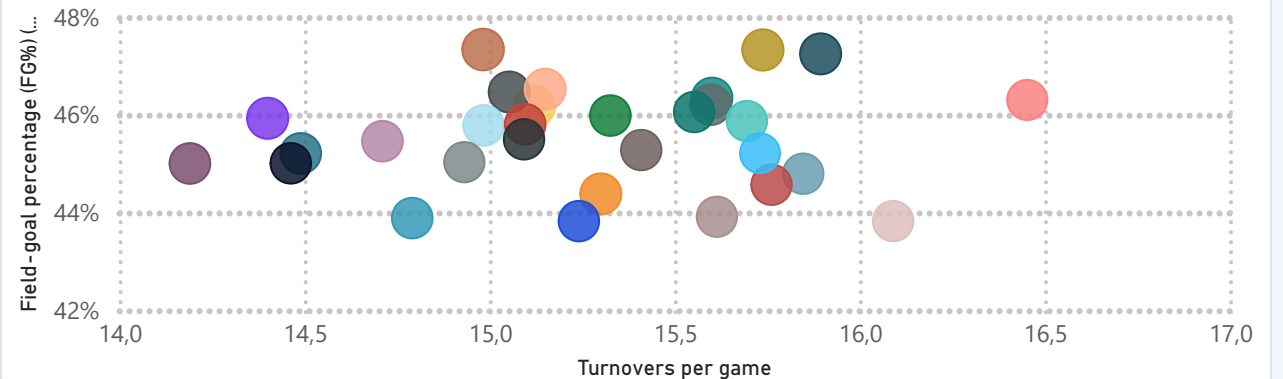
Home scoring averages 104.69 PPG versus 101.11 PPG away.

The difference is about 3.6 PPG across team-seasons; the Pelicans have the lowest win-rate CV (20.81%).

Home vs. away scoring | 1946–2022 | team-season PPG



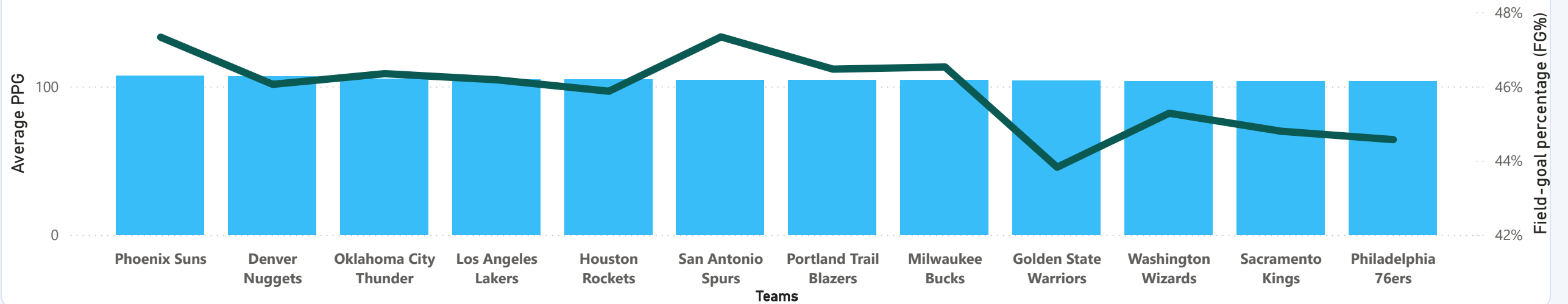
Shooting vs. turnovers | 1946–2022 | FG% and turnovers/game



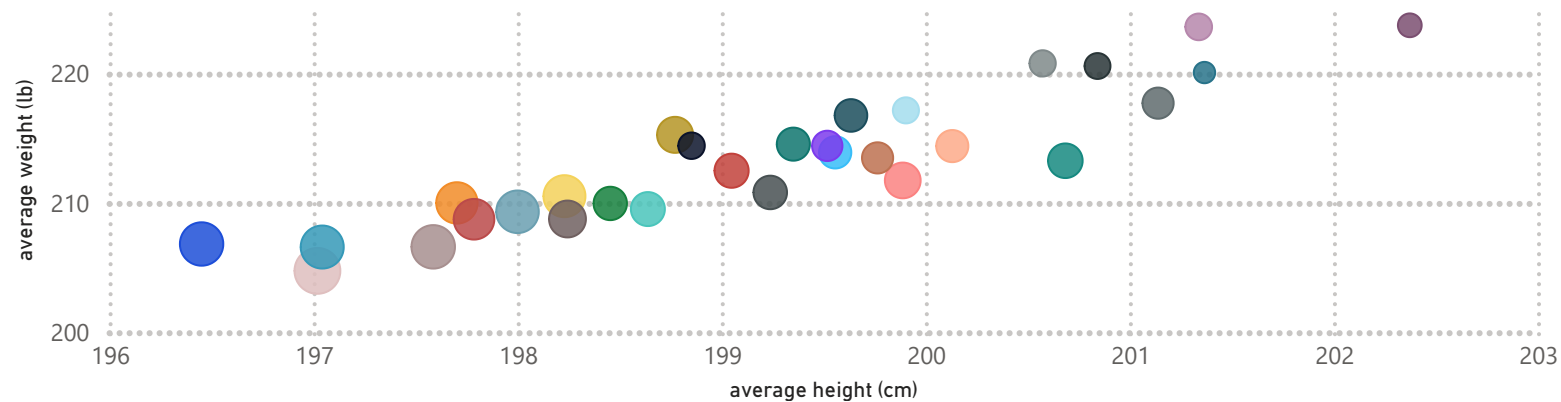


QUESTION What do offensive context and the available historical profiles show together?

Offensive context: top 12 | historical sample | PPG and FG%



Player profiles | 3,225 historical records | height (cm), weight (lb)



Team

Todas

KEY FINDING · FULL SAMPLE

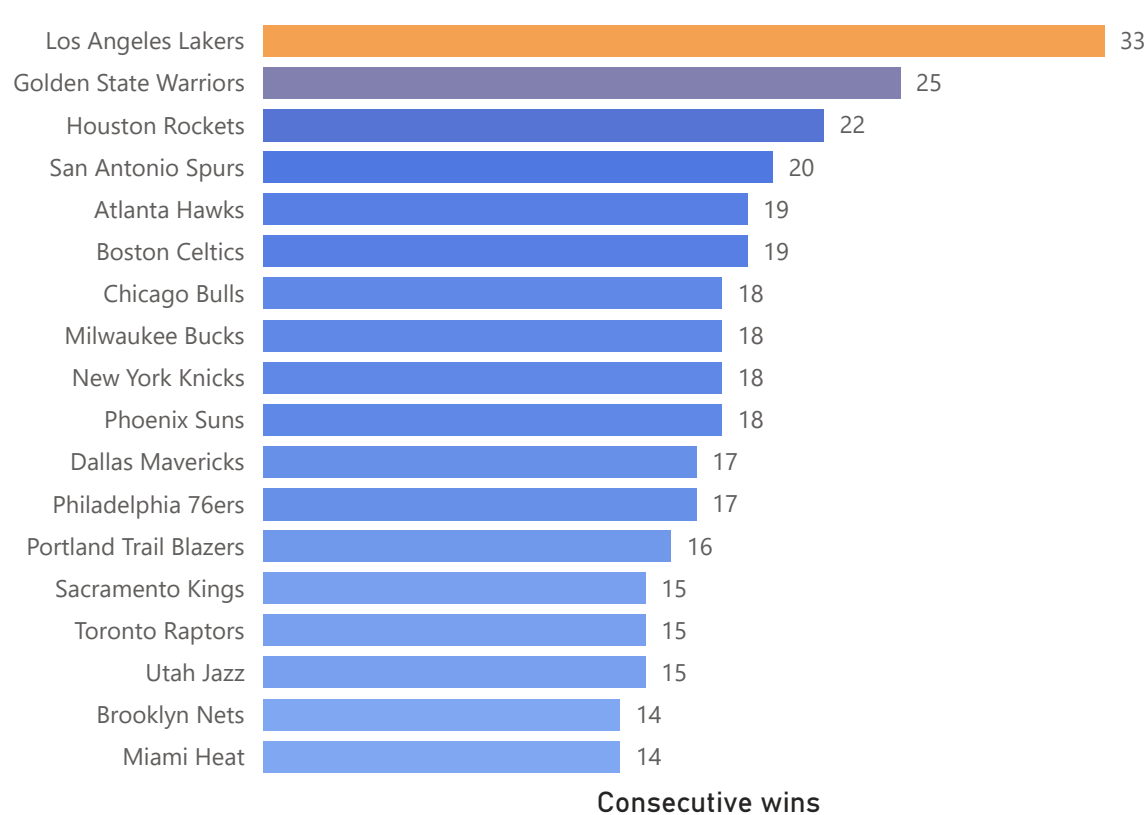
3,225 historical profiles: team-average means of 199.24 cm and 213.36 lb.

The largest team sample contains 197 historical player profiles.

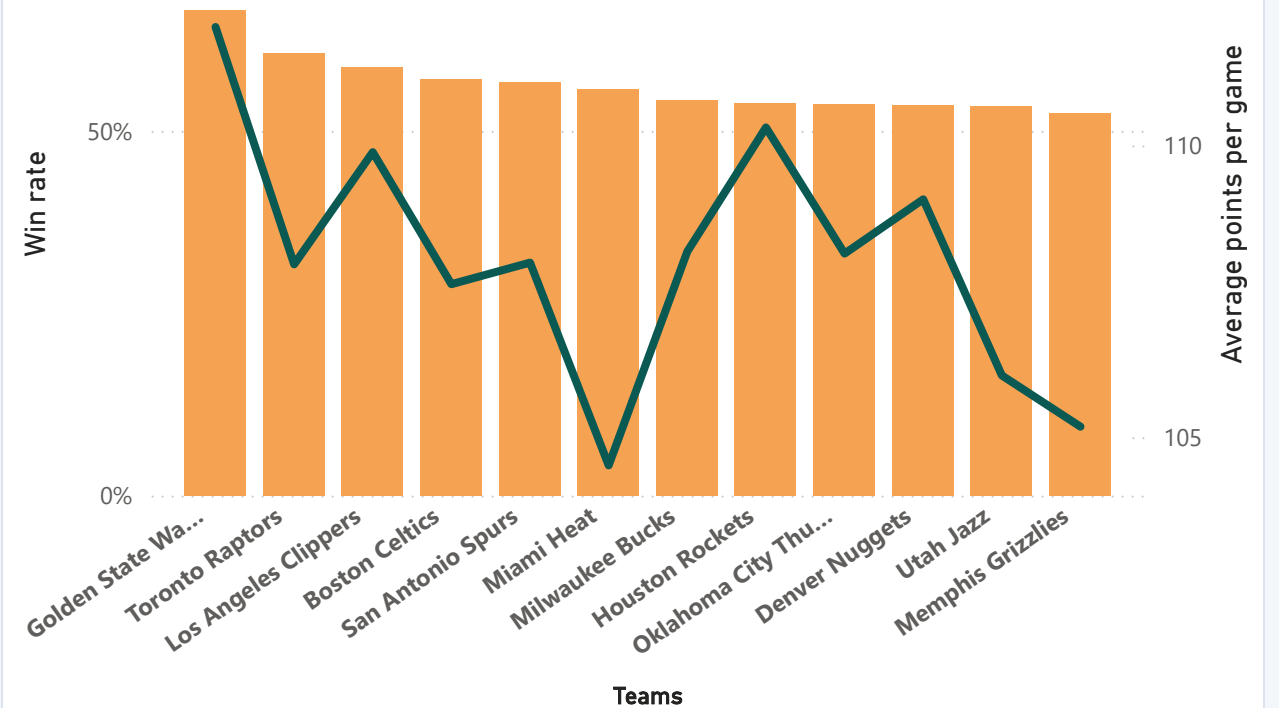


QUESTION Do the longest observed streak and the highest 2013–2022 win rate belong to the same team?

Historical peak | 1946–2022 | longest winning streak



Recent window | 2013–2022 | top 12 by win rate



KEY FINDING - FULL SAMPLE

The Lakers have the longest observed streak (33); the Warriors lead 2013–2022 at 66.53% (656/986 games).



05 · Method & Evidence

6 versioned CSVs → contract v1.0.0 → Python ETL → SQL Server → 15 analytical objects → Power BI DirectQuery

DATA

6 CSVs · 161,111 rows

65,642 unique games · seasons 1946–2022

MODEL

15 analytical objects

30 mapped teams · zero cross-table orphans

QUALITY

13 tests · 89.92% core ETL coverage

SQL reconciliation · idempotent load · schema checks

LINEAGE

SHA-256 + evidence

Manifests, result snapshot and versioned Power BI source

REVIEW & REPRODUCTION

Reproducible Python ETL · SQL Server model · reconciliation queries · compiled PBIT · pbi-tools source

A reviewer can run the ETL, load the model, reconcile the snapshot and inspect the same DirectQuery report.